

Analytical inverse kinematic solution for the redundant 7-DoF manipulator

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ABSTRACT

This paper proposes a novel parameterization analytical method to solve the inverse kinematics of a redundant 7-DoF manipulator. The current arm-angle parameterization methods cannot describe the self-motion of the elbow joint well. To address this issue, a novel arm length parameter that describes the manipulator's redundancy is defined to represent the self-motion manifold of the elbow motion. Then the analytical inverse kinematics solutions can be obtained by utilizing arm length parameter and global configuration parameters. The joint angle constraints and singularity constraints are mapped to the arm length parameter space, and the feasible arm length intervals are derived. Furthermore, a redundant parameter optimization method is proposed based on the hierarchical motion control strategy, which applies the feasible arm length interval into the redundancy resolution problem. Numerical simulations are conducted to verify the proposed inverse kinematics method can reduce motion amplitude of joint angles during trajectory tracking and thus improve the motion accuracy of the manipulator compared with the traditional methods.

1. Introduction

Compared to traditional six degrees-of-freedom (DoF) manipulators, redundant manipulators offer enhanced motion flexibility [1,2], superior obstacle avoidance capabilities [3,4], and improved fault tolerance [5]. These advantages have led to their widespread adoption across a variety of application domains, including industrial manufacturing [6,7], aerospace [8], and the medical field [9]. However, the increased number of DoFs also significantly complicates motion planning and control. In this context, inverse kinematics plays a fundamental role in enabling precise positioning and orientation of the manipulator's end-effector, as it involves computing joint angles based on a given end-effector pose.

In recent years, numerous effective approaches have been proposed to address the inverse kinematics problem for redundant manipulators, drawing considerable attention from the research community. Conventional inverse kinematics methods can generally be classified into kinematic transformation methods and differential inverse kinematics methods. The latter often suffer from several limitations, including high computational complexity and the tendency to yield only a single solution [10]. Moreover, the use of velocity Jacobian pseudo-inverses may lead to singularities [11], and joint angle constraints are typically not considered [12]. As a result, current research on inverse kinematics for redundant manipulators has increasingly focused on kinematic

transformation methods, which can be further divided into analytical and numerical approaches. Although numerical methods provide greater flexibility in handling redundancy, they are often challenged by issues such as accuracy, convergence, and a lack of analytical interpretability [13–15]. Therefore, this article primarily focuses on analytical inverse kinematics solutions for redundant manipulators.

In redundant manipulators, the ability of the manipulator to adjust its joint configurations without altering the end-effector pose is referred to as the self-motion property. Consequently, the inverse kinematics solution space of a redundant manipulator is no longer a discrete set of solutions but rather a continuous space containing infinitely many solutions. All joint variable combinations that satisfy the given end-effector pose constraints constitute the self-motion manifold. To explicitly characterize this self-motion property, researchers often introduce redundancy parameterization methods. For instance, one common approach is to define an arm angle parameter, representing the angle of a specific key joint or link relative to a designated parameter coordinate system. Shimizu et al. [12] proposed an arm angle parameterization method to describe the self-motion of non-offset SRS (Spherical–Revolute–Spherical) redundant manipulators and derived an analytical inverse kinematics solution based on this representation. This method considers joint constraints in the global configuration space and provides feasible inverse solutions. Building upon this work, subsequent

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studies [16–19] have further improved and extended the analytical solution methods. Due to limitations in operational performance, later research began to focus on the offset-type SSRMS (Space Station Remote Manipulator System) redundant manipulators. However, such manipulators often face difficulties in selecting appropriate arm angle parameters, which increases the complexity of inverse kinematics computation. To address these challenges, Xu et al. [20] proposed a dimension reduction method based on Denavit–Hartenberg (DH) parameter reconstruction, which simplifies a 7-DoF manipulator into a 6-DoF system to derive analytical inverse solutions under single-joint locking failures. Wang et al. [21] formulated the kinematic model of the SSRMS manipulator using the Product of Exponentials (POE) method. By locking one joint, the inverse kinematics problem is reduced to a 6-DoF case, and the analytical solution is obtained by solving the Paden-Kahan subproblems. Although these approaches can effectively solve inverse kinematics in specific scenarios, they may compromise the motion flexibility of the manipulator to some extent. Furthermore, Jin et al. [22] employed elbow-direction parameterization to characterize the self-motion of the SSRMS manipulator and derived an analytical inverse kinematics solution. A particle swarm optimization (PSO) algorithm was introduced to determine the optimal elbow orientation, enabling simultaneous avoidance of obstacles and singular configurations. Zhao et al. [23,24] designed two lockable passive telescopic links to enhance the reconfigurability of the SSRMS manipulator. By combining the cyclic coordinate descent method with joint angle parameterization, they derived an analytical inverse solution for the reconfigurable manipulator. Moreover, a multi-objective PSO algorithm incorporating joint torque and manipulability constraints was employed to optimize redundant parameters.

Similar to the inverse kinematics solution methods for SRS manipulators, Ma et al. [25] proposed a parameterization approach combining nominal and actual arm angles to characterize the self-motion of SSMRS manipulators. Numerical experiments demonstrated that this method achieves higher accuracy in obtaining analytical inverse kinematic solutions compared to parameterization methods based solely on joint angles. Wang et al. [26] introduced an arm angle parameterization technique for SSMRS manipulators and employed a semi-analytical approach to determine the feasible range of arm angles under singular configurations. Sinha et al. [27] proposed a dual-parameter inverse kinematics solution method for a 7-DoF manipulator with multiple joint offsets, utilizing the first joint angle and a redundant circular parameter at the elbow joint. Analytical solutions were obtained while considering joint angle constraints. In summary, the rational design of redundant parameters to describe the self-motion characteristics of redundant manipulators is of great significance for efficient inverse kinematics solutions.

Motivated by the aforementioned studies, this paper proposes a method to characterize the self-motion properties of redundant manipulators based on arm length parameterization. Firstly, the spatial self-motion behavior of the elbow joint is systematically analyzed for the designed 7-DoF manipulator. Under given end-effector poses and global configuration parameters, the feasible interval of this redundant parameter is comprehensively investigated by considering joint constraints and singularity conditions. Furthermore, an arm length optimization algorithm is developed based on a hierarchical motion control strategy, aiming to further enhance the manipulator's control accuracy.

The organization of this article is as follows: Section 2 presents the overall configuration and forward kinematics model of the designed 7-DoF manipulator; Section 3 provides an in-depth analysis of the manipulator's self-motion based on the arm length parameterization method and derives the analytical inverse kinematic solution; Section 4 explores the feasible interval of the arm length under joint limit constraints and proposes a novel redundancy resolution strategy to improve trajectory tracking accuracy; Section 5 validates the effectiveness and feasibility of the proposed analytical solution through numerical simulations; finally, Section 6 concludes the paper and discusses future work.

Table 1

Modified D-H parameters of the 7-DoF manipulator.

i	α_{i-1} (rad)	a_{i-1} (m)	d_i (m)	θ_i (rad)	θ_i limits
1	0	0	0.695	θ_1	$[-5\pi/6, 5\pi/6]$
2	$\pi/2$	0.170	0	θ_2	$[-\pi/2, \pi/2]$
3	0	1.350	0	θ_3	$[-8\pi/9, 8\pi/9]$
4	$-\pi/2$	1.150	0.140	θ_4	$[-5\pi/6, 5\pi/6]$
5	$\pi/2$	0	0.859	θ_5	$[-4\pi/3, 4\pi/3]$
6	$\pi/2$	0	0	θ_6	$[-3\pi/4, 3\pi/4]$
7	$\pi/2$	0	0	θ_7	$[-2\pi, 2\pi]$

2. Inverse kinematic solution

2.1. Manipulator configuration of the 7-DoF manipulator

The physical prototype of the designed 7-DoF manipulator is illustrated in Fig. 1(a), and its corresponding kinematic chain is depicted in Fig. 1(b). In the initial configuration, the axes of the first and fourth joints are parallel to each other, as are the axes of the second and third joints. Moreover, the axes of the first and fourth joints are orthogonal to those of the second and third joints, and these two sets of axes lie in distinct planes. The last three joints – namely the fifth, sixth, and seventh – form a spherical wrist. A notable feature of this manipulator is the incorporation of an additional redundant joint with horizontal rotation at the position of the third joint, which endows the system with positional redundancy and enhances its dexterity in spatial tasks. The kinematic parameters of the 7-DoF manipulator are listed in Table 1. The forward kinematics model is established using the modified D-H convention. The overall transformation from the base frame to the end-effector frame is expressed as:

$${}^0T = {}^0T_1T_2T_3T_4T_5T_6T_7 \quad (1)$$

where each homogeneous transformation matrix ${}^{i-1}T_i$ is defined as:

$${}^{i-1}T_i = \text{Rot}(x_{i-1}, \alpha_{i-1}) \text{Trans}(x_{i-1}, a_{i-1}) \text{Trans}(z_i, d_i) \text{Rot}(z_i, \theta_i)$$

$$= \begin{bmatrix} {}^{i-1}R & {}^{i-1}p \\ \mathbf{0} & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \cos \theta_i & -\sin \theta_i & 0 & a_{i-1} \\ \sin \theta_i \cos \alpha_{i-1} & \cos \theta_i \cos \alpha_{i-1} & -\sin \alpha_{i-1} & -d_i \sin \alpha_{i-1} \\ \sin \theta_i \sin \alpha_{i-1} & \cos \theta_i \sin \alpha_{i-1} & \cos \alpha_{i-1} & d_i \cos \alpha_{i-1} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2.2. Self-motion analysis of the manipulator

In order to facilitate solving the inverse kinematics problem, as illustrated in Fig. 1(b), the zeroth link is defined as the base, the second joint as the shoulder, the third joint as the elbow, the fifth, sixth and seventh joints form a spherical wrist, and the seventh link as the end-link. Given the presence of the spherical wrist, a commonly used decoupling principle can be applied—dividing the inverse kinematics into two subproblems: the inverse position problem and the inverse orientation problem. The joint variables $\{\theta_1, \theta_2, \theta_3, \theta_4\}$ are used to solve the inverse position problem from the base to the wrist center, while joint variables $\{\theta_5, \theta_6, \theta_7\}$ are employed to achieve the desired inverse orientation of the end-link. According to Eq. (1), the reachable spatial position p_e of the elbow joint in space can be expressed as:

$$\begin{cases} p_{ex} = {}^0T(4, 1) = (a_1 + a_2 \cos \theta_2) \cos \theta_1 \\ p_{ey} = {}^0T(4, 2) = (a_1 + a_2 \cos \theta_2) \sin \theta_1 \\ p_{ez} = {}^0T(4, 3) = a_2 \sin \theta_2 + d_1 \end{cases} \quad (2)$$

where $p_e = (p_{ex}, p_{ey}, p_{ez})^T$ represents the position of elbow joint and the homogeneous transformation matrix ${}^0T = {}^0T_2T_3T_4T_5T_6T_7$.

As illustrated in the light-colored region of Fig. 2, the reachable position of the elbow joint in space forms a complex three-dimensional

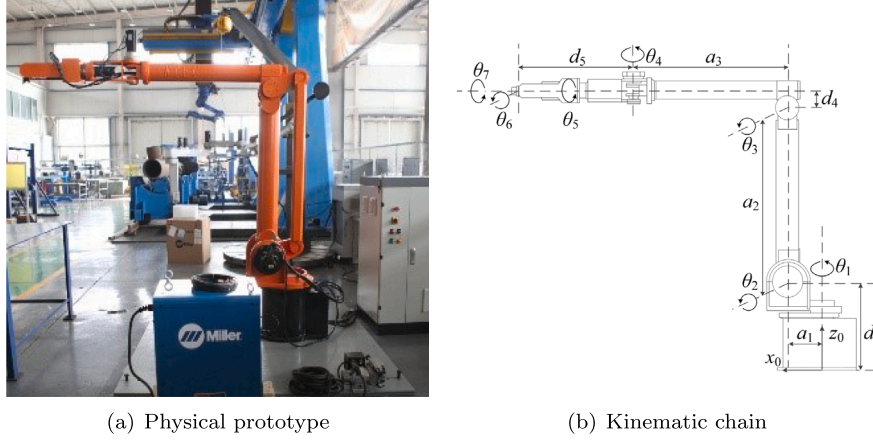


Fig. 1. Physical prototype and kinematic chain of the designed 7-Dof manipulator.

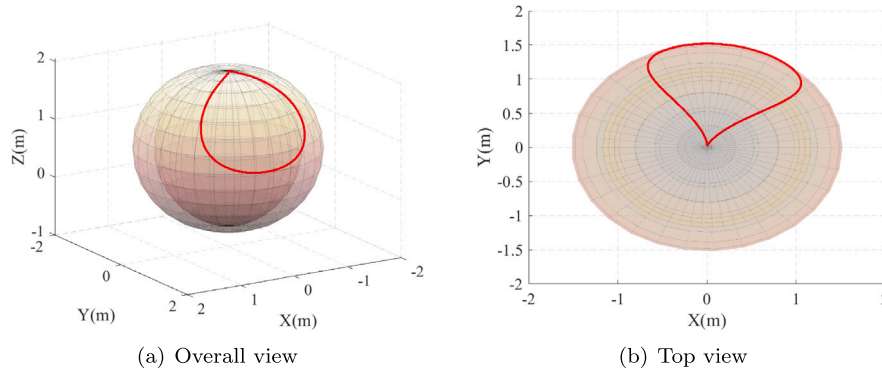


Fig. 2. The reachable position and one possible self-motion of the elbow joint.

surface, which is smoothly embedded by two ellipsoids. The red curve depicted in Fig. 2 represents a subset of the self-motion manifold of the elbow joint, with the wrist joint fixed at a specific position. It is evident that this red curve exhibits a complex self-motion trajectory. In contrast, traditional self-motion trajectories described using arm-angle parameterization methods typically form planar redundant circles, which are inadequate for characterizing the self-motion behavior of the elbow joint in the proposed manipulator configuration. To more intuitively and conveniently describe the elbow joint's self-motion behavior, the position of the wrist joint can be projected onto the plane formed by the shoulder and elbow joints, and the concept of an arm-length parameter can be introduced for characterization. As shown in Fig. 3(a), in the base coordinate system (ignoring the z -direction offset d_1), the plane formed by the shoulder and elbow joints is defined as the arm plane, which is actually perpendicular to the rotation plane of the fourth joint. Therefore, as illustrated in Fig. 3(b), the arm length ℓ is defined as the length of the projection of the wrist joint onto the arm plane. The expression for calculating the arm length ℓ is given as follows:

$$\ell = d_5 \sin \theta_4 \quad (3)$$

2.2.1. Elbow redundancy

As illustrated in Fig. 4, when the joint angle $\theta_1 = 0$, the projection point W' of the wrist joint onto the arm plane has coordinates $({}^1x_5, {}^1z_5)$ in the base coordinate frame. Under this configuration, the reachable positions of the elbow joint within the arm plane form a circle generated by the rotation about the shoulder joint. The specific location of the elbow joint in the arm plane is determined by the intersection of two circles: the first is centered at the shoulder joint with a radius of a_2 ,

representing the fixed link between the shoulder and elbow joints; the second is centered at the projection point W' with a radius of $\ell_{EW'}$, which reflects the geometric constraint between the elbow joint and the projected wrist position. The radius $\ell_{EW'}$ is calculated as follows:

$$\ell_{EW'} = \sqrt{(a_3 + \ell)^2 + d_4^2} \quad (4)$$

The value of the arm length ℓ directly determines the precise position of the elbow joint within the arm plane. As previously mentioned, the arm plane serves as a relative reference plane. Given the known arm length, the corresponding joint angle θ_1 can be calculated, which further specifies the position of the elbow joint on the spatial self-motion manifold.

2.3. Inverse kinematic computation

Given the end-effector pose matrix 0_7T and the arm length parameter ℓ , all joint angles can be determined. To address the global self-motion manifold problems in inverse kinematics, the global configuration parameters GC_k are introduced to determine the branches of the inverse kinematics uniquely. The global configuration parameters are divided into four variables that represent the sign of the joint angle coordinates at the base (GC_1), elbow (GC_2) and (GC_4), as well as wrist (GC_6). These variables affect the joint configuration of the manipulator in space.

$$GC = [GC_4 \ GC_1 \ GC_2 \ GC_6] \quad (5)$$

2.3.1. Analytical solution of θ_4

Based on the definition of the arm length in Eq. (3), the value of θ_4 can be directly obtained.

$$\theta_4 = \text{atan2}(\ell, -{}^1y_5) \quad (6)$$

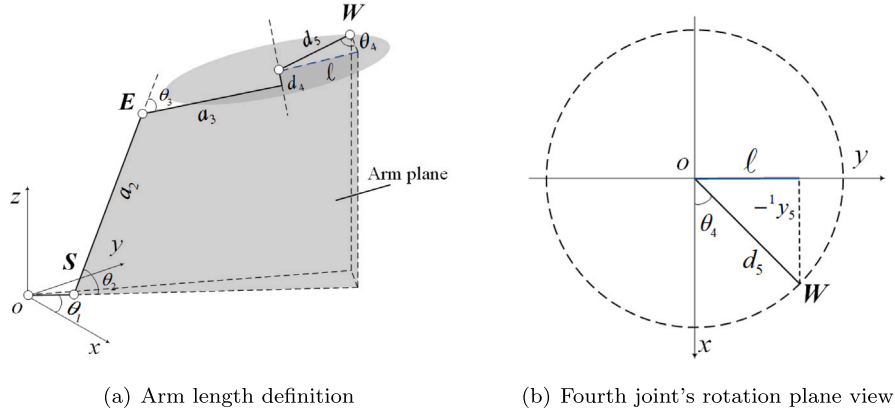


Fig. 3. The definition of arm plane and arm length.

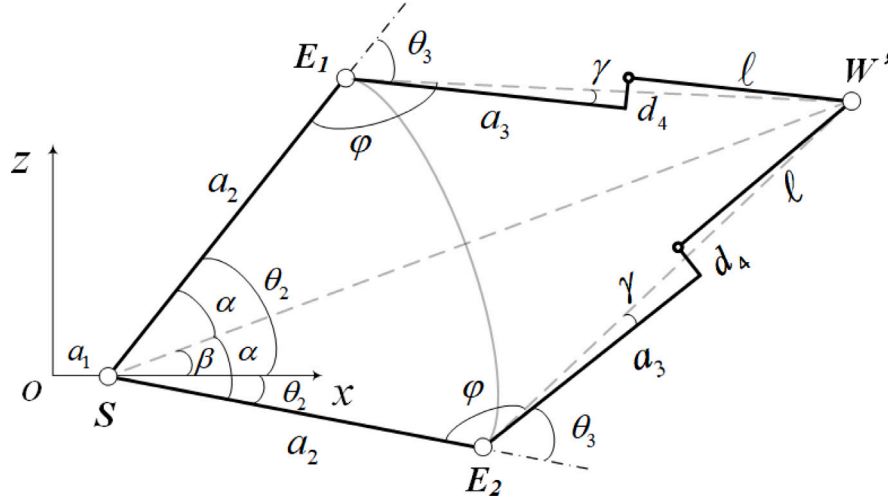


Fig. 4. Self-motion of the elbow joint in arm plane.

2.3.2. Analytical solution of θ_1

From the forward kinematics formula ${}^0_7p = {}^0_1T_5 {}^1_5p$, it can be derived that

$$\begin{cases} {}^0x_7 = {}^0x_5 = {}^1x_5 \cos \theta_1 - {}^1y_5 \sin \theta_1 \\ {}^0y_7 = {}^0y_5 = {}^1x_5 \sin \theta_1 + {}^1y_5 \cos \theta_1 \\ {}^0z_7 = {}^0z_5 = d_1 + {}^1z_5 \end{cases} \quad (7)$$

By squaring both the first and second equations in Eq. (7) and adding them together, we obtain

$${}^0y_7^2 + {}^0x_7^2 = {}^1y_5^2 + {}^1x_5^2 \quad (8)$$

As shown in Fig. 3(b), we can directly obtain 1y_5 as

$${}^1y_5 = GC_4 \sqrt{d_5^2 - \ell^2} \quad (9)$$

$$\text{where } GC_4 = \begin{cases} 1 & \text{if } {}^1y_5 \geq 0 \\ -1 & \text{if } {}^1y_5 < 0 \end{cases}.$$

By combining Eqs. (8) and (9), we can also obtain 1x_5 as

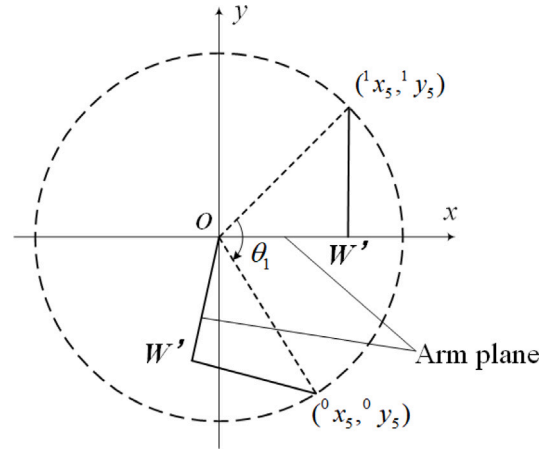
$${}^1x_5 = GC_1 \sqrt{{}^0y_7^2 + {}^0x_7^2 - {}^1y_5^2} \quad (10)$$

$$\text{where } GC_1 = \begin{cases} 1 & \text{if } {}^1x_5 \geq 0 \\ -1 & \text{if } {}^1x_5 < 0 \end{cases}.$$

Thus, the values of 1x_5 , 1y_5 and 1z_5 are all obtained.

Consequently, as depicted in Fig. 5, we can obtain the value of θ_1

$$\theta_1 = \text{atan2}({}^0y_7, {}^0x_7) - \text{atan2}({}^1y_5, {}^1x_5) \quad (11)$$

Fig. 5. The relationship between $({}^0x_5, {}^0z_5)$ and $({}^1x_5, {}^1z_5)$ in first joint's rotation plane.

2.3.3. Analytical solution of θ_2 and θ_3

In Fig. 4, the variables α and β are defined as

$$\beta = \text{atan2}({}^1z_5, {}^1x_5 - a_1) \quad (12)$$

$$\alpha = GC_2 \arccos \frac{a_2^2 + (({}^1x_5 - a_1)^2 + {}^1z_5^2) - ((a_3 + \ell)^2 + d_4^2)}{2a_2 \sqrt{({}^1x_5 - a_1)^2 + {}^1z_5^2}} \quad (13)$$

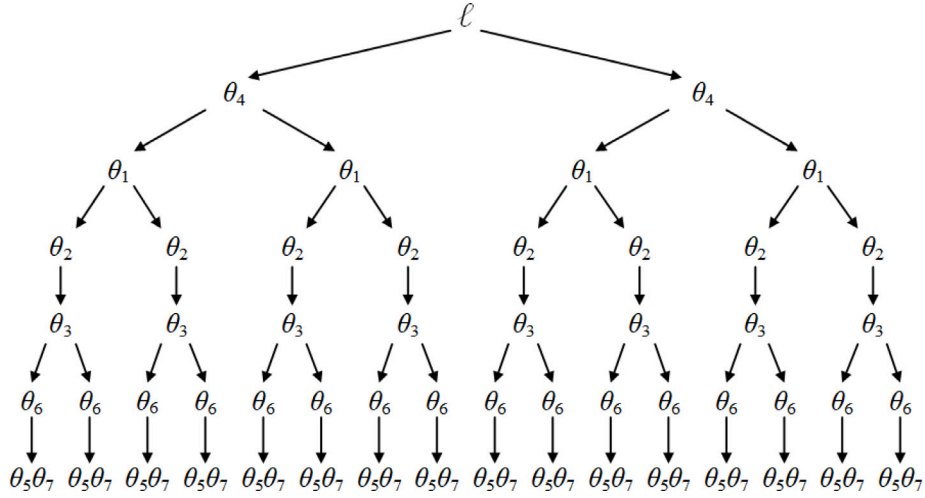


Fig. 6. Process of solving the inverse kinematics problem in the joint space.

where $GC_2 = \begin{cases} 1 & \text{if } \alpha \geq 0 \\ -1 & \text{if } \alpha < 0 \end{cases}$. Therefore, the value of θ_2 is

$$\theta_2 = \beta + \alpha \quad (14)$$

By applying the *cosine* rule, φ is calculated as

$$\varphi = \arccos \frac{\left((a_3 + \ell)^2 + d_4^2\right) + a_2^2 - \left((x_5 - a_1)^2 + z_5^2\right)}{2a_2\sqrt{(a_3 + \ell)^2 + d_4^2}} \quad (15)$$

The parameter γ in Fig. 4 is defined as

$$\gamma = \text{atan2}(d_4, a_3 + \ell) \quad (16)$$

Therefore, the value of θ_3 is

$$\theta_3 = -\gamma + GC_2(\varphi - \pi) \quad (17)$$

2.3.4. Analytical solution of θ_5 , θ_6 and θ_7

The pose matrix of wrist joint 4_7R is computed numerically using the joint angles θ_1 , θ_2 , θ_3 and θ_4 .

$$\begin{aligned} {}^4_7R &= {}^4_5R {}^5_6R {}^6_7R = ({}^1_4R {}^1_2R {}^2_3R {}^3_4R)^{-1} {}^0_7R \\ &= \begin{bmatrix} * & * & \cos \theta_5 \sin \theta_6 \\ -\cos \theta_7 \sin \theta_6 & \sin \theta_6 \sin \theta_7 & \cos \theta_6 \\ * & * & \sin \theta_5 \sin \theta_6 \end{bmatrix} \end{aligned} \quad (18)$$

Consequently, the values of θ_5 , θ_6 and θ_7 can be obtained as

$$\theta_5 = \text{atan2}({}^4_7R(3,3), {}^4_7R(1,3)) \quad (19)$$

$$\theta_6 = GC_6 \arccos({}^4_7R(2,3)) \quad (20)$$

$$\theta_7 = \text{atan2}({}^4_7R(2,2), {}^4_7R(2,1)) \quad (21)$$

where $GC_6 = \begin{cases} 1 & \text{if } \theta_6 \geq 0 \\ -1 & \text{if } \theta_6 < 0 \end{cases}$.

After solving the inverse position problem, the joint angles θ_5 , θ_6 , and θ_7 corresponding to the orientation can be subsequently determined. As illustrated in Fig. 6, a total of sixteen sets of inverse kinematics solutions exist based on the proposed arm length parameterization method.

3. Feasible inverse solutions under joint limits

To ensure stable motion of the manipulator in task space, joint limits and singularity avoidance should be considered. Therefore, a mapping method is proposed where joint limits and singularities are represented as intervals along the arm length circle. These intervals can be used to

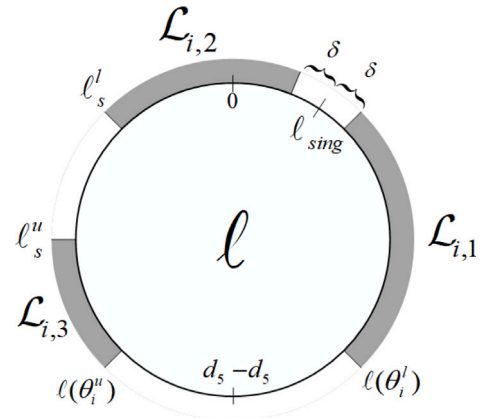


Fig. 7. Schematic diagram of feasible arm length range.

determine whether the manipulator may encounter problems such as joint limit violation, singularity violation, or even a complete lack of solutions for certain parameters (as shown in Fig. 7). ℓ_s^u and ℓ_s^l denote the upper and lower bounds of the unsolvable region for the i th joint; $\ell(\theta_i^l)$ and $\ell(\theta_i^u)$ represent the arm lengths corresponding to the lower and upper joint limits of the i th joint, respectively; and ℓ_{sing} indicates the arm length associated with a singular configuration. The feasible intervals are denoted by $\mathcal{L}_{i,j}$, where the subscript j refers to the j th feasible interval of the i th joint. The final global feasible interval $\mathcal{L}_{all}$ is obtained as the intersection of the feasible intervals for all joints, as expressed mathematically below:

$$\mathcal{L}_{all} = \bigcap_{i=1}^7 \mathcal{L}_i \quad \text{and} \quad \mathcal{L}_i = \bigcup_{j=1}^{n_j} \mathcal{L}_{i,j} \quad (22)$$

The proposed mapping method takes into account the influence of joint configuration on mapping joint limits and singularities to the redundancy space. The feasible intervals for the redundancy parameters are determined by the relationship between the joint angles and elbow self-motion, as described in Section 2. The main goal of this section is to determine the feasible arm lengths that are constrained by the joints.

3.1. Feasible arm lengths with joint 1 constraints

To establish the mapping relationship between the feasible intervals of the arm length and the constraints on joint angles, it is necessary to perform differentiation on the analytical expression relating the arm length parameter to the joint angles. Based on Eq. (11), the derivative

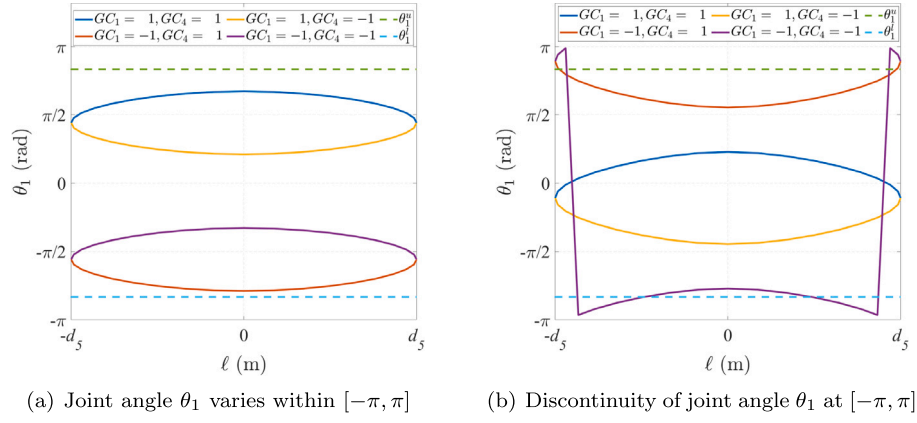


Fig. 8. Results of joint angle $\theta_1(\ell)$ when global configuration parameters are specified.

of the joint angle θ_1 with respect to the arm length parameter ℓ can be derived as follows:

$$\frac{d\theta_1}{d\ell} = -\frac{\ell}{GC_4 GC_1 \sqrt{(d_5^2 - \ell^2)(0y_7^2 + 0x_7^2 - d_5^2 + \ell^2)}} \quad (23)$$

From Eq. (23), it can be deduced that, with the specified global configuration parameters GC_1 and GC_4 , at the stationary point where $\ell = 0$, $\theta_1(\ell)$ obtains either its maximum or minimum value. Simultaneously, for $\ell = -d_5$ and $\ell = d_5$, $\theta_1(\ell)$ respectively achieves its minimum or maximum value. For example, when the global configuration parameters are set as $GC_1 = 1$ and $GC_4 = 1$, as shown by the blue curve in Fig. 8(a), $\theta_1(\ell)$ reaches its maximum value at $\ell = 0$ and its minimum values at $\ell = -d_5$ and $\ell = d_5$. By comparing the maximum and minimum value of the function $\theta_1(\ell)$ with the joint limits, we can determine the feasible intervals of arm length ℓ . Furthermore, by examining the denominator term of Eq. (23), if the position of the given target point satisfies the condition $0y_7^2 + 0x_7^2 < d_5^2$, then the feasible range of ℓ should be intersected with the interval $[0y_7^2 + 0x_7^2 - d_5^2, d_5^2 - 0y_7^2 - 0x_7^2]$.

It is worth noting that in Fig. 8(b), when the global configuration parameters are specified as $GC_1 = -1$ and $GC_4 = 1$, the purple curve is discontinuous. This is not due to kinematic singularity but because the range of the function atan2 is limited to $[-\pi, \pi]$. Therefore, when the joint angles exceeds this range, the returned angles will suddenly change from π to $-\pi$.

3.2. Feasible arm lengths with joint 2 and joint 3 constraints

According to Eq. (14), the derivative of θ_2 with respect to ℓ is given by

$$\frac{d\theta_2}{d\ell} = \frac{d\beta}{d\ell} + \frac{d\alpha}{d\ell} \quad (24)$$

where

$$\frac{d\beta}{d\ell} = \frac{{}^1z_5\ell}{{}^1x_5 \left(({}^1x_5 - a_1)^2 + {}^1z_5^2 \right)}$$

$$\begin{aligned} \frac{d\alpha}{d\ell} = & -GC_2 \left(GC_1 \ell (a_1 - {}^1x_5) \left(a_2^2 - (a_3 + \ell)^2 - d_4^2 + ({}^1x_5 - a_1)^2 + {}^1z_5^2 \right) - \right. \\ & \left. 2 \left(a_1 GC_1 \ell + a_3 \sqrt{-d_5^2 + x_7^2 + y_7^2 + \ell^2} \right) \left(({}^1x_5 - a_1)^2 + {}^1z_5^2 \right) \right) \\ & / \left(\sqrt{-d_5^2 + x_7^2 + y_7^2 + \ell^2} \left(({}^1x_5 - a_1)^2 + {}^1z_5^2 \right) \right. \\ & \left. \sqrt{4a_2^2 \left((a_3 + \ell)^2 + d_4^2 \right) - \left((a_3 + \ell)^2 + d_4^2 + a_2^2 - ({}^1x_5 - a_1)^2 - {}^1z_5^2 \right)^2} \right) \end{aligned}$$

According to Eq. (17), the derivative of θ_3 with respect to ℓ is given by

$$\frac{d\theta_3}{d\ell} = \frac{d\gamma}{d\ell} + GC_2 \frac{d\varphi}{d\ell} \quad (25)$$

where

$$\frac{d\gamma}{d\ell} = \frac{d_4}{d_4^2 + (a_3 + \ell)^2}$$

$$\frac{d\varphi}{d\ell} =$$

$$\frac{2 \left((\ell + a_3)^2 + d_4^2 \right) \left(a_3 + \frac{\ell a_1}{{}^1x_5} \right) - (\ell + a_3) \left(a_2^2 + (\ell + a_3)^2 + d_4^2 - (a_1 - {}^1x_5)^2 - {}^1z_5^2 \right)}{\sqrt{(\ell + a_3)^2 + d_4^2} \sqrt{4a_2^2 \left((\ell + a_3)^2 + d_4^2 \right) - \left(a_2^2 + (\ell + a_3)^2 + d_4^2 - (a_1 - {}^1x_5)^2 - {}^1z_5^2 \right)^2}}$$

According to Eqs. (24) and (25), it can be observed that, after the global configuration parameters GC_1 and GC_2 are specified, for any $\ell \in [-d_5, d_5]$, $\frac{d\theta_2}{d\ell}$ and $\frac{d\theta_3}{d\ell}$ are either always greater than 0 or always less than 0. This indicates that the functions $\theta_2(\ell)$ and $\theta_3(\ell)$ are monotonic and do not contain stationary points (as shown in Fig. 9). Consequently, the maximum and minimum values of $\theta_2(\ell)$ and $\theta_3(\ell)$ can only occur at the positions $\ell = -d_5$ and $\ell = d_5$.

According to Eqs. (13) and (15), the inverse cosine must be calculated when calculating joint angles θ_2 and θ_3 in Section 2.3. The boundary conditions $[\ell_s^l, \ell_s^u]$ for the existence of function values of $\theta_2(\ell)$ and $\theta_3(\ell)$ are as follows:

$$\begin{cases} \frac{a_2^2 + \left(({}^1x_5 - a_1)^2 + {}^1z_5^2 \right) - \left((a_3 + \ell_s^u)^2 + d_4^2 \right)}{2a_2 \sqrt{\left(({}^1x_5 - a_1)^2 + {}^1z_5^2 \right)}} = 1 \\ \frac{a_2^2 + \left(({}^1x_5 - a_1)^2 + {}^1z_5^2 \right) - \left((a_3 + \ell_s^l)^2 + d_4^2 \right)}{2a_2 \sqrt{\left(({}^1x_5 - a_1)^2 + {}^1z_5^2 \right)}} = -1 \end{cases} \quad (26)$$

If $\cos \alpha > 1$, the inverse kinematic solutions of some global configuration do not exist, as shown in Fig. 10. Geometrically, the triangle SEW' depicted in Fig. 4 does not exist, which is regarded as parametric singularity.

Similar to θ_1 , the discontinuity in the purple curve of θ_2 in Fig. 10(a) is also due to the limited range of the function atan2 , which is $[-\pi, \pi]$. Therefore, when the joint angles exceed this range, the returned angles will suddenly change from π to $-\pi$.

3.3. Feasible arm lengths with joints 5, 6 and 7 constraints

Based on Eqs. (19)–(21), the curves of θ_5 , θ_6 , and θ_7 with respect to the arm length parameter ℓ are shown in Figs. 11(a)–13(a) after specifying the global configuration parameters. From Fig. 13(a), it can be seen that as θ_6 approaches the wrist joint singularity point $\theta_6 = 0$, θ_6 remains continuous, and meanwhile the curves corresponding to θ_5 and θ_7 for the respective configurations are also continuous. Therefore, it is not necessary to consider singular arm lengths corresponding to the wrist joint.

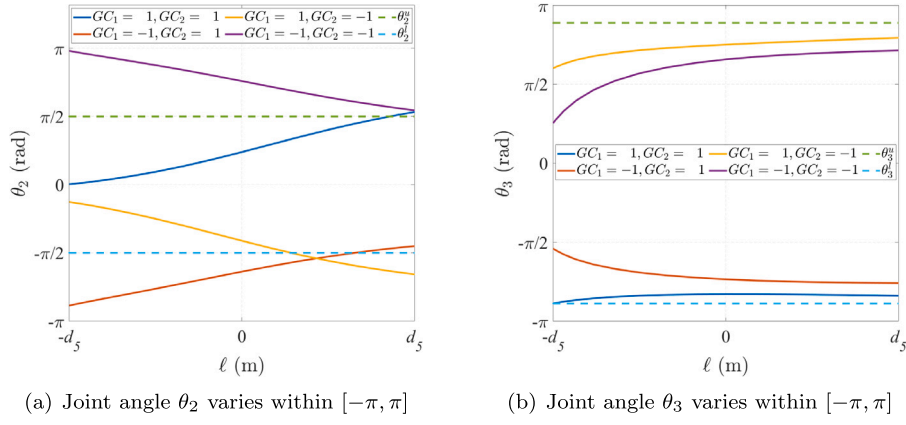


Fig. 9. Joint angles $\theta_2(\ell)$ and $\theta_3(\ell)$ after specifying the global configuration parameters.

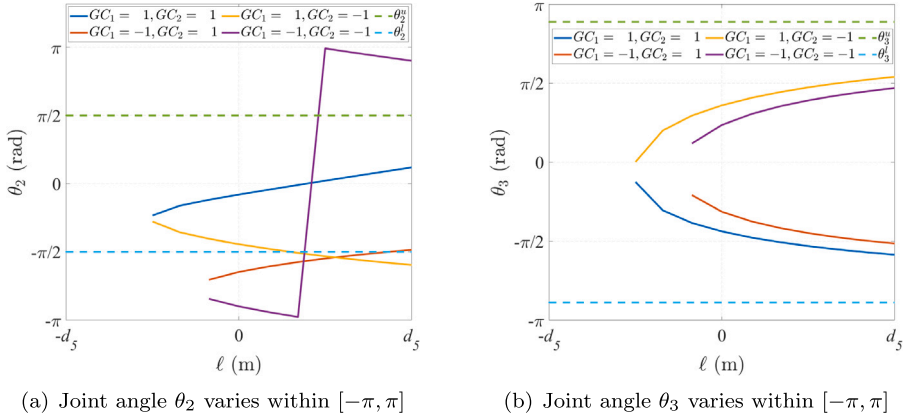


Fig. 10. Joint angles $\theta_2(\ell)$ and $\theta_3(\ell)$ varies when unsolvable.

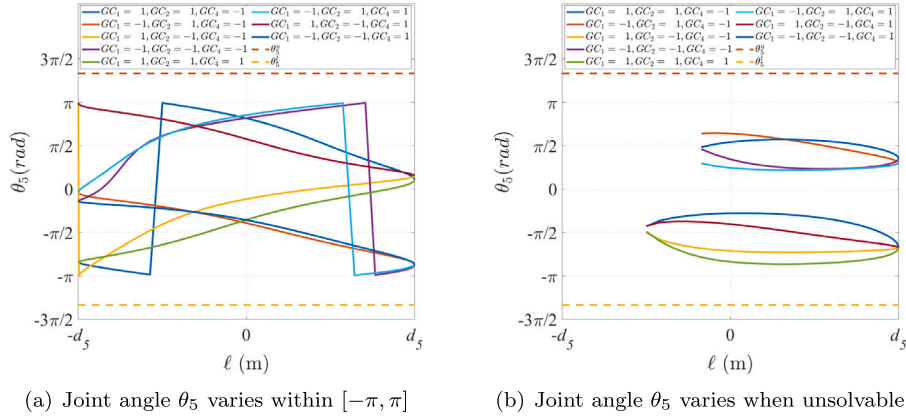


Fig. 11. Joint angles $\theta_5(\ell)$ after specifying the global configuration parameters.

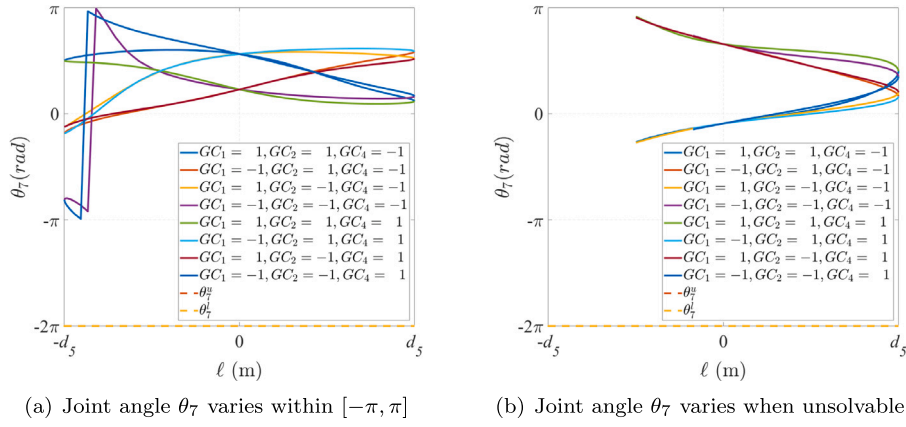
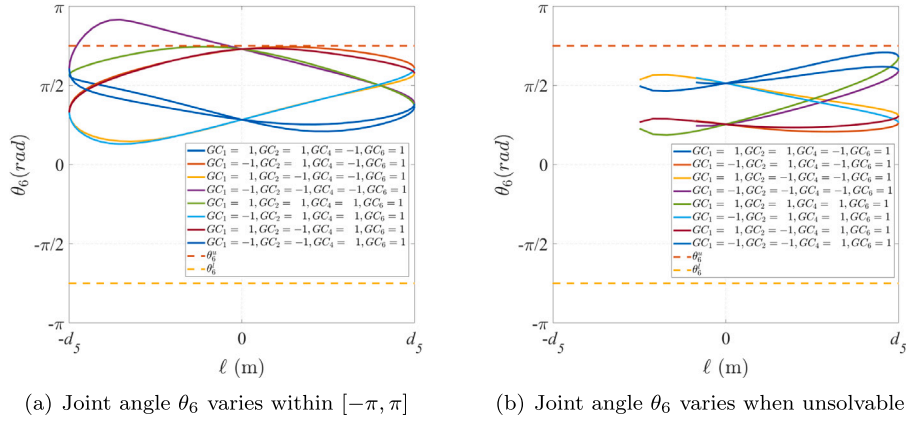
Similar to θ_1 , due to the presence of a reversal point, the curve of θ_7 for certain configurations in Fig. 12(a) exhibits a sudden change from π to $-\pi$. Furthermore, since the values of $\theta_5 - \theta_7$ are computed based on θ_2 and θ_3 , when the critical condition in Eq. (26) is not satisfied, as illustrated in Figs. 11(b)–13(b), θ_5 , θ_6 , and θ_7 share the same interval $[\ell_s^l, \ell_s^u]$ with θ_2 and θ_3 .

4. Redundancy resolution

Under specific operating conditions, by constraining the angular variation $\Delta\theta_1$ of the rotational base joint and prioritizing the motion control of joint θ_4 , the positioning error of the end-effector can be

effectively reduced, thereby enabling the desired operation of the manipulator. Based on this premise, this paper proposes a redundancy resolution algorithm that adopts a hierarchical motion control strategy. As described in Section 3, joint limits and singularities can be mapped onto an arm-length circle via the feasible arm-length interval, thus incorporating them uniformly into the motion planning constraints. Based on Eq. (27), a simple metric-based arm length update method is further proposed. Specifically, given the feasible arm-length interval $\mathcal{L}_{all}$ at the current time, the arm length ℓ_t is updated from the previous time step ℓ_{t-1} by solving the following optimization problem:

$$\underset{\ell_t}{\operatorname{argmin}} \quad \Delta\theta_1(\ell_t) \quad \text{s.t.} \quad \ell_t \in \mathcal{L}_{all}. \quad (27)$$

Fig. 12. Joint angles $\theta_7(\ell)$ after specifying the global configuration parameters.Fig. 13. Joint angles $\theta_6(\ell)$ after specifying the global configuration parameters.

Taking a total differential on Eq. (11), the cost function $\Delta\theta_1$ is defined as

$$\Delta\theta_1(\ell_i) = \left| \frac{\partial\theta}{\partial^0x_7} \Delta^0x_7 + \frac{\partial\theta}{\partial^0y_7} \Delta^0y_7 + \frac{\partial\theta}{\partial\ell} (\ell_i - \ell_{i-1}) \right| \quad (28)$$

where the partial derivatives in the above expression are given as:

$$\begin{cases} \frac{\partial\theta}{\partial^0x_7} = -\frac{{}^0y_7}{({}^0x_7^2 + {}^0y_7^2)} - \frac{GC_1 GC_4 {}^0x_7 \sqrt{-\ell^2 + d_5^2}}{({}^0x_7^2 + {}^0y_7^2) \sqrt{\ell^2 - d_5^2 + {}^0x_7^2 + {}^0y_7^2}} \\ \frac{\partial\theta}{\partial^0y_7} = \frac{{}^0x_7}{({}^0x_7^2 + {}^0y_7^2)} - \frac{GC_1 GC_4 {}^0y_7 \sqrt{-\ell^2 + d_5^2}}{({}^0x_7^2 + {}^0y_7^2) \sqrt{\ell^2 - d_5^2 + {}^0x_7^2 + {}^0y_7^2}} \end{cases}$$

These partial derivatives can be numerically computed using the Euler method at the operating point ℓ_i . The pseudocode of the corresponding redundant resolution algorithm is presented in Algorithm 1.

5. Simulation works

5.1. The closed-form solution validation

Simulation experiments are conducted to evaluate the performance of the proposed analytical inverse kinematic solution for the 7-DoF redundant manipulator. The simulations are carried out on a computer running the Windows 11 operating system, equipped with an Intel(R) Core(TM) i5-12500H processor (3.10 GHz) and 16.0 GB of RAM. The simulation environment is implemented using MATLAB software. The arm length parameter is randomly assigned as $\ell = -0.7803$ m, and the

Algorithm 1: Redundancy resolution of the 7-DoF manipulator

Input: End-effector trajectory 0T_i , global configuration parameter $[GC_k]$, initial arm length ℓ_0

Output: Joint angles sequences $\{\theta_{1...7_i}\}$

```

1 for  $i = 1$  to  $\text{size}({}^0T_i)$  do
2    $\mathcal{L}_{\text{all}} \leftarrow$  computing feasible intervals of  ${}^0T_i$ ;
3   while  $\mathcal{L}_{\text{all}}$  does not exist do
4      $[GC_k] \leftarrow [GC_k]_{\text{new}}$ ;
5      $\mathcal{L}_{\text{all}} \leftarrow$  computing feasible intervals of  ${}^0T_i$ ;
6    $\ell_i \leftarrow \min(\Delta\theta_1(\ell_i, \ell_{i-1}))$  s.t.  $\ell_i \in \mathcal{L}_{\text{all}}$ ;
7    $\ell_{i-1} \leftarrow \ell_i$ ;
8    $\theta_{1...7_i} \leftarrow$  solving inverse kinematics using  ${}^0T_i$ ,  $\ell_i$ , and  $[GC_k]$ ;

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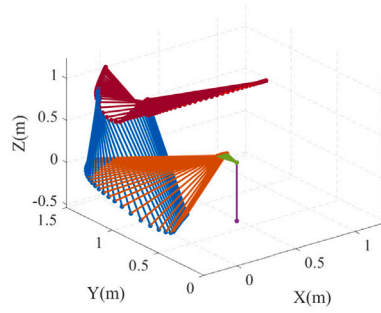
desired pose is defined by the following matrix:

$${}^0_7T = \begin{bmatrix} -0.4143 & -0.3493 & -0.8405 & 0.2359 \\ 0.3942 & -0.9012 & 0.1802 & 1.2758 \\ -0.8203 & -0.2567 & 0.5110 & 0.4729 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (29)$$

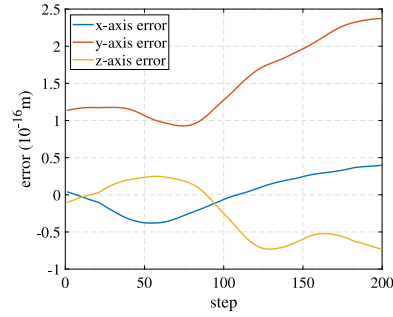
As shown in Table 2, sixteen sets of analytical solutions are derived, along with the corresponding global configuration parameters $[GC_k]$. From Table 2, it can be seen that the proposed inverse kinematics algorithm based on arm length parameterization can effectively solve the inverse kinematics problem of the designed redundant manipulators.

Table 2
Sixteen sets of analytical solutions for 7-DoF redundant manipulator.

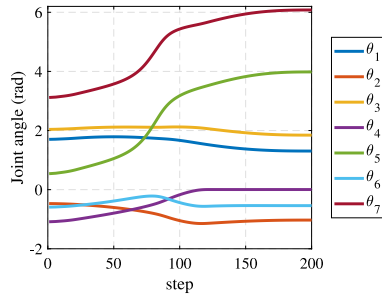
GC	θ_1 (rad)	θ_2 (rad)	θ_3 (rad)	θ_4 (rad)	θ_5 (rad)	θ_6 (rad)	θ_7 (rad)
[1 1 1 1]	1.1074	0.0482	-2.7388	-2.0022	-2.5019	1.9552	1.5267
[1 1 1 -1]	1.1074	0.0482	-2.7388	-2.0022	0.6398	-1.9552	4.6683
[1 1 -1 1]	1.1074	-0.4551	2.0149	-2.0022	2.9182	1.4566	-0.2079
[1 1 -1 -1]	1.1074	-0.4551	2.0149	-2.0022	-0.2234	-1.4566	2.9338
[1 -1 1 1]	-1.4731	-2.7075	-1.8601	-2.0022	0.3436	0.5808	-0.2992
[1 -1 1 -1]	-1.4731	-2.7075	-1.8601	-2.0022	-2.7981	-0.5808	2.8424
[1 -1 -1 1]	-1.4731	3.0186	1.1361	-2.0022	-0.5162	1.4965	-3.1066
[1 -1 -1 -1]	-1.4731	3.0186	1.1361	-2.0022	2.6254	-1.4965	0.0349
[-1 1 1 1]	1.6685	0.0482	-2.7388	-1.1393	-2.7872	1.6444	1.6988
[-1 1 1 -1]	1.6685	0.0482	-2.7388	-1.1393	0.3544	-1.6444	-1.4429
[-1 1 -1 1]	1.6685	-0.4551	2.0149	-1.1393	-2.6914	0.6279	-0.1018
[-1 1 -1 -1]	1.6685	-0.4551	2.0149	-1.1393	0.4503	-0.6279	3.0399
[-1 -1 1 1]	-2.0342	-2.7075	-1.8601	-1.1393	-0.2911	1.4876	-0.3260
[-1 -1 1 -1]	-2.0342	-2.7075	-1.8601	-1.1393	2.8505	-1.4876	2.8157
[-1 -1 -1 1]	-2.0342	3.0186	1.1361	-1.1393	-0.1663	2.5775	-2.7031
[-1 -1 -1 -1]	-2.0342	3.0186	1.1361	-1.1393	2.9753	-2.5775	0.4385



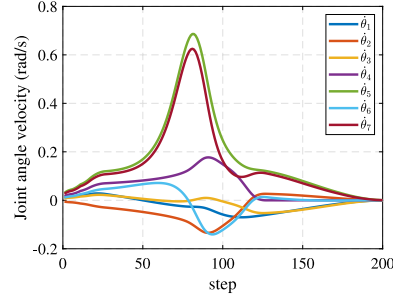
(a) Trajectory tracking in Cartesian space



(b) Error of inverse kinematics calculation



(c) Joint angle tracking curve



(d) Joint velocity tracking curve

Fig. 14. Simulation results for inverse kinematics of linear trajectories.

5.2. Trajectory tracking optimization validation

To further validate the effectiveness of the proposed redundancy resolution algorithm, a simulation experiment of linear trajectory tracking for the redundant manipulator is conducted. In this experiment, the end-effector trajectory is generated using a quintic polynomial planning method, with the initial pose matrix of the trajectory consistent with that defined in Eq. (29). Under this initial pose, after calculating the feasible range of arm length, a random initial value $\ell_0 = -0.7803$ m is selected. The global configuration parameter is set as

$$GC = [GC_4 \ GC_1 \ GC_2 \ GC_6] = \begin{bmatrix} -1 & 1 & -1 & -1 \end{bmatrix}.$$

The pose matrix at the endpoint of the trajectory is defined as

$${}^0T^e = \begin{bmatrix} -0.4143 & -0.3493 & -0.8405 & 1.2359 \\ 0.3942 & -0.9012 & 0.1802 & 1.2758 \\ -0.8203 & -0.2567 & 0.5110 & 0.4729 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

That is, the trajectory is a straight-line motion along the X -axis with a total length of 1.000 m, during which the end-effector orientation remains unchanged throughout the entire movement.

The simulation results illustrate the manipulator's trajectory tracking performance in Cartesian space, as shown in Fig. 14(a). The motion curves and joint velocity variation curves of each joint during trajectory tracking are shown in Fig. 14(c) and Fig. 14(d) respectively. The corresponding positional errors are shown in Fig. 14(b), with all error values kept below 1×10^{-15} . It is important to note that such extremely small errors arise from numerical round-off in computer calculations rather than reflecting practical tracking accuracy or methodological limitations. These results verify that the proposed analytical inverse kinematics algorithm exhibits high computational precision, thereby demonstrating its effectiveness for applications requiring high accuracy. To further evaluate the algorithm's performance, twenty experiments are conducted. The average total processing time for trajectory tracking is 1132.76 ms, corresponding to an average iteration time of 5.66 ms per redundancy resolution step, demonstrating that the proposed method possesses favorable real-time capabilities.

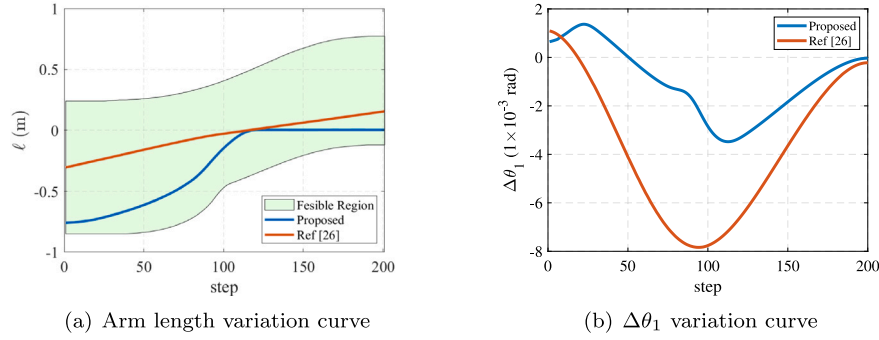
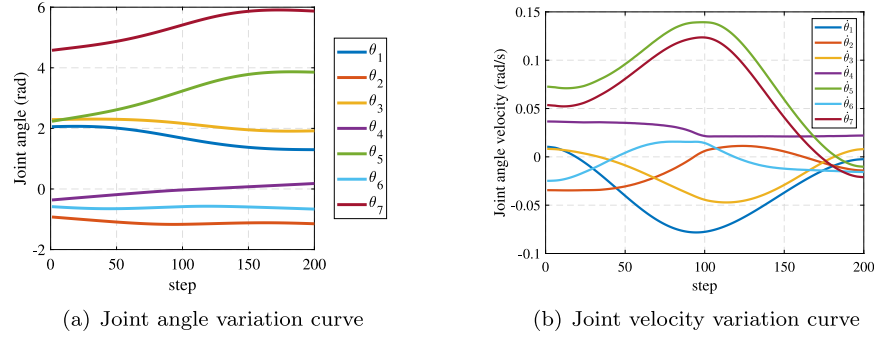
Fig. 15. Arm length and $\Delta\theta_1$ variation curve.

Fig. 16. Variation curve of the joint angle and velocity using traditional method.

During trajectory tracking, the feasible range of arm lengths is obtained as illustrated in Fig. 15(a). The optimal arm length using the hierarchical motion control strategy is also obtained as the solution to the one-dimensional optimization problem, depicted by the blue curve in Fig. 15(a). The red curve in Fig. 15(a) represents the arm length computed using the redundant parameter update strategy (Eq. (30)) based on the traditional inverse S-curve ($K = 0.005$, $\lambda = 10$) in Ref. [26].

$$\psi_t = \psi_{t-1} + K \left(\frac{\psi_{\max} - \psi_{\min}}{1 + e^{\lambda(\psi_{t-1} - \psi_{\min})}} - \frac{\psi_{\max} - \psi_{\min}}{1 + e^{-\lambda(\psi_{t-1} - \psi_{\max})}} \right) \quad (30)$$

Fig. 16(a) illustrates the variation of joint angles during the trajectory tracking process using the redundant parameter update strategy based on the conventional inverse S-curve, while Fig. 16(b) shows the corresponding joint velocity variation curve. Compared with the traditional redundant parameter optimization method, which tends to drive the arm length toward the midpoint of the feasible range during tracking, the proposed algorithm not only maintains the validity of solutions within the feasible arm length range but also effectively suppresses the motion amplitude of the base joint. Furthermore, as shown in Fig. 15(b), the proposed method significantly reduces the motion amplitude of the revolute joint θ_1 . Specifically, the angular variation of joint θ_1 is 22.65° using the proposed method, in contrast to 43.46° with the traditional approach. These results verify the superior performance of the proposed analytical redundancy resolution method in reducing joint motion amplitudes.

6. Conclusion

This paper proposes a novel parameterization analytical method to solve the inverse kinematics of a redundant 7-DoF manipulator. Due to the intricate self-motion characteristics of the elbow joint based on arm-angle parameterization methods, a novel arm length parameter that describes the manipulator's redundancy is defined to represent the self-motion manifold of the elbow motion. The analytical inverse kinematics solutions can be obtained by utilizing arm length parameters and global configuration parameters. Simulation works demonstrate

that the inverse kinematic solving algorithm exhibits high computational accuracy and real-time performance. Subsequently, joint angle constraints and singularity constraints are mapped into the arm length parameter space to deduce feasible intervals for the arm length. A novel redundant parameter optimization method is proposed based on a hierarchical motion control strategy, incorporating the feasible arm length intervals into the redundant redundancy problem. Simulation results indicate that, compared to the traditional inverse S-curve strategy, the proposed algorithm can reduce the motion amplitude of the joint θ_1 during trajectory tracking, thereby enhancing the motion precision of the manipulator.

CRedit authorship contribution statement

Gang Wang: Supervision, Project administration, Methodology, Conceptualization. **Wenjun Li:** Writing – original draft, Methodology, Investigation, Formal analysis. **Xiaoshan Gao:** Writing – review & editing, Validation, Funding acquisition. **Qi Zhang:** Visualization, Software, Resources, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

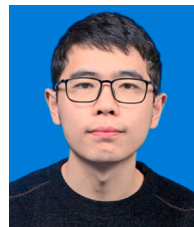
No data was used for the research described in the article.

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